Correlation Regression

Regression lines of y on x. $y = a + bx$. — (1)

Let $y_i = a + bx_i$

$\therefore \sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} x_i$ — (1) $\Rightarrow \bar{y} = a + b\bar{x}$ — (2)

and $\sum_i x_i y_i = a\sum x_i + b\sum x_i^2 \Rightarrow \frac{1}{n}\sum x_i y_i = a\bar{x} + \frac{b}{n}\sum x_i^2$ — (3)

$\therefore$ the line of regression y on x pass through $\bar{x}, \bar{y}$. Now $Cov(x,y) = \frac{1}{n}\sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n}\sum_i x_i y_i - \bar{x}\bar{y}$

$\Rightarrow \frac{1}{n}\sum_i x_i y_i = Cov(x,y) + \bar{x}\bar{y}$ — (4)

and $\sigma_x^2 = \frac{1}{n}\sum x_i^2 - \bar{x}^2 \Rightarrow \frac{1}{n}\sum x_i^2 = \sigma_x^2 + \bar{x}^2$ — (5)

$\therefore$ from (3) we have $Cov(x,y) + \bar{x}\bar{y} = a\bar{x} + b(\sigma_x^2 + \bar{x}^2)$

multiplying (2) by $\bar{x}$ and subtracting from (6) we have

$Cov(x,y) = b\sigma_x^2 \Rightarrow b = \frac{Cov(x,y)}{\sigma_x^2}$.

Since b is the slope of eqn. (1) and it passes through $(\bar{x}, \bar{y})$ $\therefore$ eqn. of regression line of y on x is $y - \bar{y} = b(x - \bar{x}) \Rightarrow y - \bar{y} = \frac{Cov(x,y)}{\sigma_x^2}(x - \bar{x})$.

$\Rightarrow y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})$. $\quad \because r = \frac{Cov(x,y)}{\sigma_x \sigma_y}$

$Cov(x,y) =$ $\quad b_{yx} = r\frac{\sigma_y}{\sigma_x}$ $\quad b_{xy} = r\frac{\sigma_x}{\sigma_y}$. R.C. y on x $=$ Cov(